

Roll No.

21N501 /

Total No. of Pages.: 03

B.Tech. II Sem (New Scheme) Main Exam Acad Session 2022-23

All Branch

(2FY1-01) - Engineering Mathematics II

Time : 3 Hours

Maximum Marks: 70

Min. Passing Marks:

Instructions to Candidates:

Part – A: Short answer questions (up to 25 words) 10×2 marks = 20 marks. All 10 questions are compulsory.

Part – B: Analytical/Problem Solving questions 5×4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving/ Design questions 3×10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1 _____

2 _____

PART-A

Q.1. Evaluate $\left[\left(\frac{-9}{2} \right) \right]$. [2]

Q.2. Evaluate $\int_0^1 \int_0^{1-x} xy dx dy$ [2]

Q.3. Evaluate $\int_0^2 (4-x^2)^{\frac{3}{2}} dx$. [2]

Q.4. Evaluate $\frac{d}{dt} \left(\vec{r} \times \frac{d\vec{r}}{dt} \right)$ [2]

Q.5. Find the directional derivative of [2]

$\theta(x, y, z) = (x^2 - 2y^2 + 4z)$ at $(1, 1, -1)$ in the direction of the vector $2\vec{i} + \vec{j} - \vec{k}$.

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Q.6. State Stoke's theorem. [2]

Q.7. Define right circular cylinder. [2]

Q.8. Find the radius of the circle [2]

$$x^2 + y^2 + z^2 - 2y - 4z = 11, x + 2y + 2z = 15.$$

Q.9. Find the eigen values of the matrix. [2]

$$A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

Q.10. Are the following vectors linearly independent [2]

$$x_1 = (1, 3, 4, 2), x_2 = (1, 2, 3, 4), x_3 = (2, 3, 4, 9)$$

PART-B

Q.1. Evaluate by changing the order of integration. [4]

$$\int_0^{\infty} \int_1^{\infty} \frac{e^{-xy}}{y} dx dy.$$

Q.2. Show that [4]

$$\int_0^2 x(8-x^3)^{1/3} dx = \frac{16\pi}{9\sqrt{3}}$$

Q.3. if $\vec{r} = xi + yj + zk$ and $r = |\vec{r}|$ then find $\nabla^2 r^n$ [4]

Q.4. Use Green's theorem to evaluate the line integral $\int_C [(xy + y^2)dx + x^2 dy]$

Where C is the boundary of the closed region bounded by $y = x$ and $y = x^2$ [4]

Q.5. Find the equation of the sphere having the circle [4]

$$x^2 + y^2 + z^2 = 9, x - 2y + 2z = 5 \text{ as a great circle.}$$

Q.6. Determine the rank of matrix [4]

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Q.7. Investigate for consistency of the following equations and if possible, Find the solutions: [4]

$$4x - 2y + 6z = 8; x + y - 3z = -1; 15x - 3y + 9z = 21$$

PART-C

- Q.1. Find the volume and surface area of the solid formed by revolution of the cardioid $r = a(1 + \cos\theta)$ about the initial line. [10]
- Q.2. Use Stoke's theorem to evaluate $\int_c [(x+2y)dx + (x-z)dy + (y-z)dz]$ [10]
Where c is the boundary of the triangle with vertices $(2, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 6)$ oriented in the anti-clockwise direction.
- Q.3. Find the equation of the sphere which passes through the circle $x^2 + y^2 = 4, z = 0$ and is cut by the plane $x + 2y + 2z = 0$ in a circle of radius 3. [10]
- Q.4. Find the eigen values and eigen vectors of the matrix [10]
$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & -6 \\ 2 & -2 & 3 \end{bmatrix}$$
- Q.5. Find the values of the constants a, b, c so that the directional derivative of $\phi = ax^2y^2 + byz + cz^2x^2$ at $P(1, 2, -1)$ has maximum magnitude 64 in the direction parallel to z -axis. [10]

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